

Q10: An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

Introduction

Image quality plays an important role in object detection. Two important concepts that affect image quality are **sampling** and **image formation**. Sampling determines the number of pixels used to represent an image, while image formation describes how light from a real-world scene is captured by a camera sensor. Poor sampling and improper image formation reduce object visibility and make object detection difficult.

Effect of Sampling on Object Visibility

Sampling is the process of converting a continuous image into a digital image by representing it with pixels.

Low Sampling Rate (Low Resolution)

- Fewer pixels are used to represent the image.
- Fine details are lost.
- Small objects become blurred or disappear.
- Object boundaries become unclear.
- The image may exhibit aliasing and pixelation.

High Sampling Rate (High Resolution)

- More pixels are used to capture the image.
- Fine details are preserved.
- Objects appear sharper and more distinct.
- Small objects can be detected accurately.

Effect of Image Formation on Object Visibility

Image formation is the process in which light reflected from objects passes through the camera lens and reaches the image sensor to form a digital image.

Factors Affecting Image Formation

1. Lighting Conditions

- Poor lighting reduces the amount of light reaching the sensor.
- Images become dark and noisy.
- Objects become difficult to identify.

2. Lens Quality

- Poor-quality lenses introduce blur and distortion.
- Object edges become unclear.

3. Sensor Quality

- Low-quality sensors produce noisy images with reduced detail.
- High-quality sensors capture clearer images.

4. Focus

- Incorrect focus produces blurred images.
- Fine object details are lost.

Impact on Object Detection

When both sampling and image formation are poor:

- Small objects cannot be recognized.
- Edges become unclear.
- Important image details are lost.
- Detection algorithms produce inaccurate results.
- Object recognition accuracy decreases.

Methods to Improve Object Visibility

1. Increase the sampling rate by using a higher-resolution camera.
2. Improve illumination during image capture.

3. Use high-quality camera lenses and image sensors.
4. Apply image enhancement techniques such as contrast enhancement and sharpening.
5. Reduce image noise using suitable filtering methods.

Applications

- Autonomous vehicles
- Surveillance systems
- Medical imaging
- Satellite image analysis
- Industrial inspection

Conclusion

Sampling and image formation together determine the visibility of objects in a digital image. A **low sampling rate** reduces image resolution and causes loss of detail, while **poor image formation** introduces blur, noise, and distortion. By increasing the sampling rate, improving lighting, using better cameras, and applying image enhancement techniques, object visibility and detection accuracy can be significantly improved.